From Subwords to Pixels: Evaluating Text Representations for Arabic Machine Translation

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Abstract

Subword tokenization underpins modern neural NLP but can be suboptimal for morphologically rich languages with substantial orthographic variation. In this paper, we compare vocabulary-based and vocabulary-free representations for Arabic-to-English machine translation, including subword, byte-level, and pixel-level approaches. Across 35 evaluation sets spanning Modern Standard Arabic (MSA) and 18 dialects, all approaches achieve comparable translation performance with sufficient supervision. While subword models perform best with limited MSA pretraining data, differences diminish with scale and dialect-specific fine-tuning. Pixel-based models further exhibit the greatest robustness to unseen orthographic variation and provide the highest training and inference throughput. We make our code and models publicly available.[1]

1 Introduction

Subword tokenization has become the de facto standard in neural natural language processing (NLP) (Sennrich et al., 2016; Kudo and Richardson, 2018; Mielke et al., 2021). However, it may be suboptimal for morphologically rich and underrepresented languages (Rust et al., 2021; Pfeiffer et al., 2021; Arnett and Bergen, 2025). In Arabic, this challenge is further compounded by the coexistence of Modern Standard Arabic (MSA) and numerous Dialectal Arabic (DA) varieties, which exhibit substantial orthographic, morphological, syntactic, and lexical variation. Under standard subword tokenization, such variation often leads to severe overtokenization, resulting in longer input sequences, reduced compression rates, and fragmented representations. These issues can ultimately affect both the efficiency and effectiveness of downstream applications, including machine translation (MT).

To address these limitations, prior work has explored both vocabulary-based and vocabulary-free alternatives to standard subword tokenization. Within the vocabulary-based paradigm, approaches such as linguistically informed tokenization and vocabulary expansion seek to better align tokenization with linguistic structure and improve vocabulary coverage (Tawfik et al., 2019; Liang et al., 2023; Asgari et al., 2025). Recently, vocabulary-free representations have emerged as a promising alternative. These include byte-level models, which operate on raw bytes, and pixel-based approaches, which encode text as images and eliminate the need for tokenization (Salesky et al., 2021; Rust et al., 2023; Lotz et al., 2025; Kesen et al., 2025). Despite these advances, the merits of vocabulary-based and vocabulary-free representations remain poorly understood for Arabic MT. In particular, it is unclear how these approaches compare in translation performance under MSA and dialect-specific supervision (**RQ1**), how they respond to varying amounts of MSA pretraining data (**RQ2**), how robust they are to orthographic variation (**RQ3**), and what computational trade-offs they exhibit during training and inference (**RQ4**).

To answer these questions, we conduct the first systematic comparison of subword, byte-level, and pixel-based representations for Arabic-to-English MT. We evaluate these approaches across 35 evaluation sets spanning MSA and 18 dialects from six benchmarks. Our results reveal trade-offs across representations. Subword models perform best in lower-data settings, but the gap narrows with more MSA pretraining data, with all approaches reaching comparable performance at full scale. Dialect-specific fine-tuning further reduces these differences, even with limited MSA pretraining. Beyond translation performance, pixel-based models are the most robust to diacritized and dotless Arabic and provide the highest training and inference throughput. Our contributions are as follows:

[1]https://github.com/mbzuai-nlp/arapixel-mt

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1. Across 35 evaluation sets spanning MSA and 18 dialects, we find comparable translation performance across representations with sufficient supervision.
2. We study the effect of MSA pretraining data size, finding that representation differences diminish with more data and DA fine-tuning.
3. We demonstrate the robustness of pixel-based models to diacritized and dotless Arabic.
4. We identify efficiency trade-offs, with pixel-based models providing the highest training and inference throughput.

2 Background and Related Work

2.1 Text Representations

Vocabulary-Based Representations Modern NLP systems represent text using fixed vocabularies learned through subword tokenization algorithms such as Byte-Pair Encoding (BPE) (Sennrich et al., 2016), WordPiece (Devlin et al., 2019), and SentencePiece (Kudo and Richardson, 2018). These methods learn tokenizations from corpus frequency statistics rather than morphological structure. Consequently, they may overtokenize morphologically complex languages, degrading performance. Prior work has addressed these limitations through vocabulary expansion, morphology-aware tokenization, and token replacement (Wang et al., 2020; Chau et al., 2020; Minixhofer et al., 2022; Kajiura et al., 2023; Dobler and de Melo, 2023; Cui et al., 2024; Bari et al., 2025; Chizhov et al., 2024; Lian et al., 2025; Jabbar, 2024; Asgari et al., 2025; Hamdan et al., 2025). Nevertheless, these approaches remain constrained by fixed vocabularies.

Vocabulary-Free Representations Vocabulary-free approaches eliminate the need for a predefined token inventory, primarily through byte-level and pixel-level representations. Byte-level models represent text as sequences of raw bytes over a fixed 256-symbol alphabet (Xue et al., 2022; Yu et al., 2023; Kallini et al., 2025; Pagnoni et al., 2025), providing complete vocabulary coverage across languages and scripts. However, because Unicode characters may span multiple bytes, byte-level models produce substantially longer sequences, particularly for non-Latin scripts such as Arabic (Arnett et al., 2024; Unicode Consortium, 2025). Pixel-based approaches instead process visually rendered text, and have been shown to be robust to orthographic variation (Rust et al., 2023), effective for multilingual-to-English MT including MSA (Salesky et al., 2021, 2023), and competitive on multilingual language understanding benchmarks (Kesen et al., 2025). Most relevantly, Lotz et al. (2025) showed that pixel-based encoding outperforms both vocabulary expansion and byte-level representations for MT in several non-Latin scripts, while enabling positive transfer across languages sharing a writing system.

Motivated by these findings, we conduct the first systematic comparison of vocabulary-based and vocabulary-free representations for Arabic MT across MSA and DA varieties.

2.2 Arabic Linguistic Challenges

Morphological Richness and Orthographic Ambiguity Arabic has a templatic and concatenative morphological system that inflects for many features while attaching clitics such as conjunctions, prepositions, and pronouns (Habash, 2010). This results in a large number of surface forms, many of which occur infrequently (Gerz et al., 2018; Botev et al., 2022). Moreover, Arabic is orthographically ambiguous because it uses optional diacritics to specify short vowels, which are typically omitted.

Dialectal Variation and Orthographic Noise Arabic is diglossic (Ferguson, 1959), with MSA coexisting alongside numerous dialectal varieties that differ in phonology, morphology, syntax, and lexicon. Unlike MSA, which has a standardized orthography, dialects are primarily spoken and lack writing conventions, leading to substantial spelling variation (Habash et al., 2018). Beyond dialectal variation, Arabic text exhibits substantial orthographic variation, including MSA spelling errors (Zaghouani et al., 2014), romanized transliteration (Darwish, 2014; Bies et al., 2014), and more recently, dotless Arabic text (Rom and Bar, 2021).

Together, these unique challenges make Arabic a particularly informative case study for understanding how different text representations handle morphological complexity and orthographic variation.

2.3 Arabic Machine Translation

Arabic MT has historically focused on MSA, with growing interest in DA in recent years. This has resulted in numerous parallel corpora and benchmarks, covering both MSA and a wide range of DA varieties (Zbib et al., 2012; Sajjad et al., 2013; Salloum and Habash, 2013; Bouamor et al., 2014;

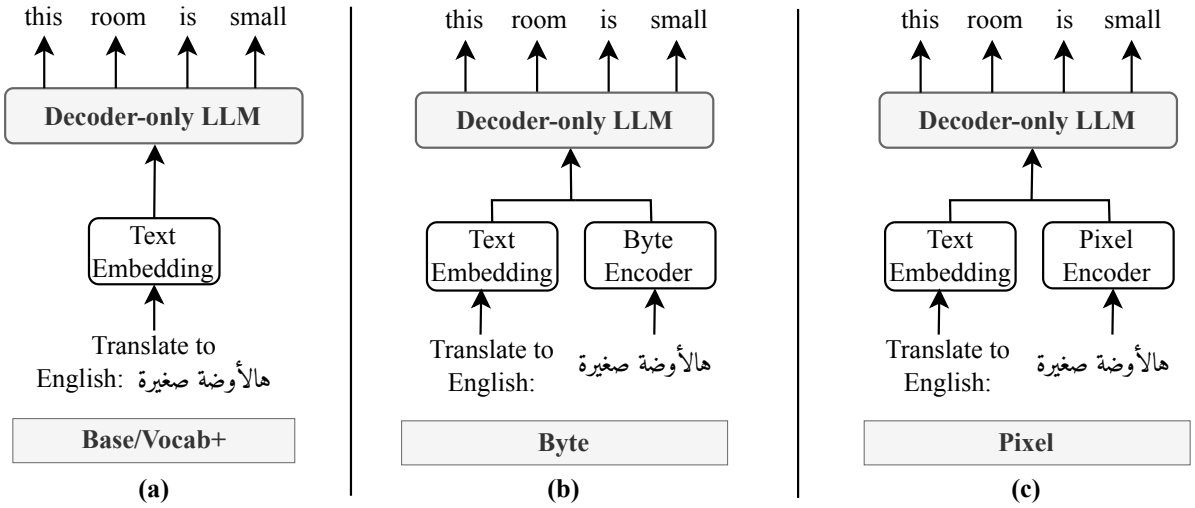


Figure 1: Illustration of the four text representation variants: Base (a), Vocab+ (a), Byte (b), and Pixel (c). The subword models (Base and Vocab+) encode the prompt and source text through the text embedding layer, whereas the byte and pixel models encode the source text using dedicated fallback encoders and combine these representations with prompt embeddings before decoding.

Elmahdy et al., 2014; Meftouh et al., 2015; Habash et al., 2017; Bouamor et al., 2018; Guzmán et al., 2019; Sajjad et al., 2020; Goyal et al., 2022; Abdul-Mageed et al., 2024; Robinson et al., 2025; Mekki et al., 2026, *inter alia*).

Modern Arabic neural MT systems predominantly rely on subword tokenization (Sajjad et al., 2020; Nagoudi et al., 2022; Kadaoui et al., 2023; Robinson et al., 2025; Jon et al., 2026). Despite the central role of text representations in these systems, relatively little work has examined their impact on Arabic MT performance. Oudah et al. (2019) and Gaser et al. (2023) investigated morphology-aware tokenization for statistical and neural MT. More recently, Salesky et al. (2021, 2023) demonstrated the effectiveness of pixel-based representations for Arabic MT within multilingual settings. However, the relative merits of vocabulary-based and vocabulary-free representations remain poorly understood, and no systematic comparison of subword, byte-level, and pixel-based approaches has been conducted for Arabic MT. Our work addresses this gap through a controlled comparison across a diverse collection of MSA and DA benchmarks.

3 Methodology

In this comparative study of Arabic-to-English MT, we fix the translation backbone to a pretrained decoder-only English language model and vary how the Arabic source text is represented and encoded. We compare two paradigms: vocabulary-based and vocabulary-free representations. The vocabulary-based paradigm includes the default subword tokenizer of the backbone model (**Base**) and a vocabulary-expanded variant (**Vocab+**), while the vocabulary-free paradigm comprises byte-level (**Byte**) and pixel-based (**Pixel**) encodings implemented through a learned hypernetwork.

Figure 1 illustrates the four representation variants considered in this work. The following input template is used across all approaches:

```
Translate to English. Arabic:
⟨source⟩ English: ⟨target⟩
```

3.1 Vocabulary-Based Representations

Our baseline (**Base**) uses the language model's original tokenizer and embedding layer. The translation prompt, Arabic source sentence, and English target are represented using the model's native subword vocabulary and processed autoregressively by the decoder-only language model.

As a stronger vocabulary-based baseline, we experiment with a standard vocabulary expansion (**Vocab+**), which augments the original vocabulary with Arabic-specific subword units. We train a new BPE tokenizer from scratch on the Arabic source side of the training data using a target vocabulary size of 32,000 tokens. Tokens that are not already present in the original tokenizer and contain at least one non-ASCII character are added to the base vocabulary, yielding an expanded vocabulary $\mathcal{V}^{+} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{new}}$.

3.2 Vocabulary-Free Encoding via Hypernetwork

Our two vocabulary-free approaches, byte- (**Byte**) and pixel-level (**Pixel**) encodings, follow the hypernetwork paradigm of Lotz et al. (2025), in which the Arabic source text bypasses the language model's standard embedding layer entirely. Instead, the source text is first segmented into words by whitespace-and-punctuation tokenization, and each word is independently encoded by a dedicated encoder into a single contextualization-free embedding vector. The resulting sequence of word embeddings is then concatenated with the standard subword token embeddings of the translation prompt and English target. The full input sequence is passed as input to the decoder-only language model, which performs cross-word contextualization and autoregressively generates the English translation.

Formally, both vocabulary-free encoders share the same interface: given a source sentence, they produce a sequence of word-level representations $\mathbf{y}_i \in \mathbb{R}^{d_{\text{enc}}}$. A projection multi-layer perceptron (MLP) maps each $\mathbf{y}_i$ to the hidden dimension d of the language model. The projected embeddings are directly substituted for the source-side subword embeddings, requiring no architectural modification to the base language model. Following Lotz et al. (2025), we do not add any special tokens at the boundary between text handled by the language model's standard embedding layer and text handled by the hypernetwork encoder.

3.3 Pixel-Based Encoding

Arabic Text Rendering A key design decision is the rendering of Arabic source words into pixel patches. Arabic presents several considerations that distinguish its rendering from Latin-script languages: right-to-left directionality, mandatory character joining and context-dependent glyph shaping (initial, medial, final, and isolated forms), optional diacritics, and the presence of ligatures. Character shaping is handled automatically by the Pango rendering engine, which produces contextually correct joined glyph forms.

Each word is rendered independently onto a canvas of $H = 24$ pixels and $W = 384$ using the GoNotoCurrent font at size 10pt (approximately 16.67px at 120 DPI) on a plain white background, with NotoSans and NotoSansArabic as fallback fonts. Images are rendered in grayscale and divided into 24×24-pixel bigram patches with one character-overlap, following Lotz et al. (2025). Words exceeding the canvas width are truncated to 384 pixels, while shorter words are padded to produce the fixed canvas size.

Encoder Architecture The pixel encoder is a Vision Transformer (ViT; Dosovitskiy et al. 2021) adapted for text, following Rust et al. (2023). Each rendered word image is divided into $N = 16$ patches, represented as $\mathbf{x} \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where $P = 24$ is the patch size and $C = 3$ is the number of image channels. The patches are then linearly projected into embeddings:

$$\mathbf{z} = \mathbf{x}\mathbf{E} + \mathbf{E}_{\text{pos}},$$

where $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times d}$ is a convolutional projection and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{N \times d}$ are 1D sinusoidal positional embeddings, reset at each word boundary. The patch embeddings are processed by a stack of transformer layers to build word representations.

Each word is encoded independently, producing a non-contextualized representation at the word level. To process multiple words efficiently in a forward pass, the batch of sentences is flattened into a single dimension, treating words across sentences as items in one batch. As a result, self-attention operates only over patches belonging to the same word, without requiring an attention mask.

The patch encodings for each word are aggregated via mean pooling to yield a single word-level representation $\mathbf{y}_i \in \mathbb{R}^{d_{\text{enc}}}$, which is then passed to the projection MLP described above. This produces a single embedding per word that is subsequently passed to the language model decoder.

3.4 Byte-Level Encoding

The byte-level encoder follows the same hypernetwork architecture as the pixel encoder, replacing rendered images with raw UTF-8 byte sequences. Each Arabic word is encoded as a sequence of up to 32 bytes (corresponding to up to 16 Arabic characters, since standard UTF-8 encodes each Arabic character as two bytes); bytes beyond this limit are truncated. Padding is handled with a dedicated pad token, giving a vocabulary of 257 symbols.

The encoder architecture mirrors the pixel encoder: byte token embeddings are augmented with 1D sinusoidal positional embeddings, then processed by a transformer-based encoder. The token representations are aggregated via masked mean pooling to produce a word-level representation, $\mathbf{y}_i \in \mathbb{R}^{d_{\text{enc}}}$, followed by the projection MLP.

Dataset	Variety	Train	Dev	Test
Alexandria	DA + MSA	66K	12K	14K
APT	DA	172K	2K	2K
Bible	DA + MSA	-	2K	2K
FLORES	DA + MSA	-	9K	9K
MADAR	DA + MSA	54K	6K	12K
OPUS	MSA	13M	-	-
QAraC	DA + MSA	-	7K	7K
QED	MSA	376K	-	-
TED	MSA	174K	-	-
UN	MSA	10M	-	-
Total		24M	38K	47K

Table 1: Corpus statistics of MT datasets.

4 Experimental Setup

4.1 Data

We use a diverse collection of Arabic-English MT datasets spanning both MSA and DA. Our evaluation suite comprises six benchmarks: Alexandria (Mekki et al., 2026), the Arabic Dialect/English Parallel Text (APT) (Zbib et al., 2012), the Bible, FLORES (Goyal et al., 2022), MADAR (Bouamor et al., 2018), and QAraC (Elmahdy et al., 2014). These benchmarks cover MSA and 18 dialectal varieties from multiple countries and regions of the Arab world. Additionally, for MSA pretraining (§4.3), we use large-scale MSA parallel corpora from OPUS (Lison and Tiedemann, 2016), QED (Guzman et al., 2013), TED (Cettolo, 2016), and the UN corpus (Ziemski et al., 2016).

All Arabic text is whitespace-and-punctuation tokenized, diacritics are removed, and Alif, Ya, and Ta Marbuta normalization is applied using CAMeL Tools (Obeid et al., 2020). We further convert Indic numerals to their Arabic counterparts. For computational efficiency, we retain only MSA pretraining sentence pairs whose Arabic source contains at most 32 words. For APT, Bible, and QAraC, we use the training (Train), development (Dev), and testing (Test) splits created by Sajjad et al. (2020); for all other datasets, we use their official publicly available splits.

Overall, the resulting collection contains 24M, 38K, and 47K sentence pairs for Train, Dev, and Test, respectively. Corpus statistics are summarized in Table 1; a detailed dialectal breakdown is provided in Table 9 (Appendix D).

4.2 Implementation Details

All systems described in §3 are built on the same decoder-only language model backbone, SmolLM2-360M-Instruct (Allal et al., 2025), whose word embedding dimension is $d = 960$. For a controlled comparison between vocabulary-free representations, the **Byte** and **Pixel** encoders share the same architecture configuration. Both use hidden size $d_{\text{enc}} = 1{,}536$, four transformer layers, 16 attention heads, and feed-forward dimension 6,144. The projection MLP consists of two layers with an intermediate dimension of 3,072. Model parameter counts are provided in Table 8 (Appendix A).

4.3 Training Setup

MSA Pretraining To compare text representations under MSA supervision (**RQ1**), we pretrain all models on the same collection of MSA-English parallel corpora: OPUS, QED, TED, and UN. **Base** is trained end-to-end for two epochs. For **Vocab+**, **Byte**, and **Pixel**, we first perform one alignment epoch, followed by two epochs of end-to-end training. For **Vocab+**, the alignment epoch freezes the original model weights and updates only the newly added embedding rows. For **Byte** and **Pixel**, we freeze the language-model backbone and train only the newly initialized encoder and projection MLP. We use a maximum input length of 256 subword tokens for **Base** and **Vocab+**, and 32 words for **Byte** and **Pixel**, covering all retained MSA training examples without truncation.

DA Fine-Tuning We further fine-tune the MSA-pretrained models on the training sets of Alexandria, APT, and MADAR to evaluate them under dialect-specific supervision (**RQ1**). We consider four settings: DA-only and mixed DA+MSA fine-tuning with DA:MSA ratios of 1:1, 1:2, and 1:4. Mixing MSA with DA aims to mitigate overfitting to the limited DA data while preserving the broader knowledge acquired during MSA pretraining. For the mixed settings, MSA examples are sampled from both the original MSA pretraining corpora and the MSA portions of Alexandria and MADAR. All models are fine-tuned for one epoch, with maximum input lengths of 256 subword tokens for **Base** and **Vocab+**, and 64 words for **Byte** and **Pixel**, covering all fine-tuning examples without truncation.

Data Scaling To study the effect of MSA pretraining data size (**RQ2**), we additionally pretrain all models on 5%, 10%, 25%, and 50% of the

MSA data, alongside the full-data (100%) setting described above. We then fine-tune each MSA-pretrained model on the DA+MSA$_{1:1}$ setting, using a balanced mixture that incorporates MSA supervision without outweighing the DA data. We retain the full DA training data at each scale, while drawing the MSA examples from the corresponding MSA pretraining subset. This allows us to examine how the MSA pretraining data size affects each representation both before and after DA fine-tuning.

Optimization Details All experiments use four A100 GPUs and a random seed of 42. We use a cosine learning rate schedule with 10% linear warmup and a peak learning rate of 3×10^{-4}. For MSA pretraining, per-device microbatch sizes are 16, 16, 32, and 64, with gradient accumulation steps of 8, 8, 4, and 2 for **Base**, **Vocab+**, **Byte**, and **Pixel**, respectively. For DA fine-tuning, the corresponding microbatch sizes are 8, 8, 8, and 16, with accumulation steps of 4, 4, 4, and 2. At inference, all models use a batch size of 32.

4.4 Evaluation Setup

We use BLEU (Papineni et al., 2002) as our primary metric and additionally report chrF++ (Popović, 2015, 2017) and COMET (Rei et al., 2020). BLEU and chrF++ are computed using SacreBLEU (Post, 2018), while COMET is computed using the official implementation.

Robustness In addition to evaluating models on the original Arabic input, we assess robustness to orthographic variation using diacritized and dotless versions of the evaluation sets (**RQ3**). Diacritized text is generated automatically using the CAMeL Tools (Obeid et al., 2020) BERT morphological disambiguator (Inoue et al., 2022). For MSA, we use the SAMA database (Graff et al., 2009), while for Egyptian and all other dialectal varieties we use CALIMA (Habash et al., 2012). Although automatically generated analyzers are available for Gulf (Khalifa et al., 2020) and Levantine (Eskander et al., 2013), they tend to be lower quality compared to CALIMA. Dotless text is generated by removing all letter dots.

Efficiency To assess computational efficiency (**RQ4**), we measure training and inference throughput in sentences per second, along with peak GPU memory usage in GiB. For training, **Pixel** inputs are rendered and cached beforehand, so the reported throughput excludes the one-time rendering cost. At inference, **Pixel** inputs are rendered online; we therefore report both end-to-end throughput, including preprocessing (rendering for **Pixel**), and preprocessing throughput separately. Training efficiency is evaluated on four A100 GPUs over 30 steps (15,360 examples), while inference efficiency is evaluated on a single A100 GPU using 3,200 Dev examples. Each experiment is repeated three times, and we report the mean and standard deviation across runs.

4.5 LLM Baselines

We compare against LLMs as a strong upper-bound baseline. Our goal is not to outperform these models, but rather to contextualize the performance of different text representation paradigms relative to contemporary general-purpose LLMs. Specifically, we evaluate two commercial models: GPT-5.5 (Singh et al., 2026) and Gemini-3.1-Flash (Team et al., 2025), as well as two Arabic-centric models: Jais-2-70B-Chat (Anwar et al., 2026) and Fanar-2-27B-Instruct (TEAM et al., 2026). For benchmarks with available training data, we use 5-shot prompting, where examples are sampled uniformly at random from the corresponding dialect of the benchmark. For benchmarks without training splits, we use zero-shot prompting. Unlike our trained models, which operate on normalized Arabic text, we provide the input Arabic text sentences to the LLMs without Alif/Ya/Ta-Marbuta normalization. The prompt we use is provided in Appendix B.

5 Results

5.1 RQ1: Effect of Text Representation

MSA Pretraining Table 2 presents Dev results after MSA pretraining. The four representations achieve comparable performance, with **Base** and **Pixel** both reaching 20.0 average BLEU, followed by **Vocab+** (19.8) and **Byte** (19.3). Paired bootstrap resampling (Koehn, 2004) shows no significant difference between **Base** and **Pixel**, while their differences with **Vocab+** and **Byte** are statistically significant. Across benchmarks, **Base** performs best on Alexandria, APT, Bible, MADAR, and QAraC, while **Pixel** performs best on FLORES. Detailed results across dialects and metrics are provided in Table 10 (Appendix E.1). These findings indicate that, with sufficient MSA pretraining, the representation approaches achieve broadly similar translation performance despite small but statistically significant differences among some models.

Corpus	Base	Vocab+	Byte	Pixel	GPT-5.5	Gemini	Jais	Fanar
Alexandria	**16.4**	**16.4**	16.0	**16.4**	43.1	43.3	42.1	39.1
APT	**10.0**	9.0	9.1	9.3	20.4	19.9	17.3	16.9
Bible	**10.6**	10.3	9.2	10.3	24.9	30.1	25.3	26.2
FLORES	29.9	29.5	29.1	**30.3**	42.3	42.0	44.1	38.5
MADAR	**23.2**	22.9	22.5	22.9	52.0	51.9	71.7	43.1
QAraC	**9.8**	9.4	9.6	9.3	17.9	19.3	18.9	15.7
Overall	**20.0**	19.8	19.3	**20.0**	40.2	40.8	43.8	36.0

Table 2: Macro-average BLEU scores on the Dev sets. Bold indicates the best MSA-pretrained model; underline indicates the best LLM for each dataset.

Data Mixture	Base	Vocab+	Byte	Pixel
MSA	**20.0**	19.8	19.3	**20.0**
+DA	**30.9**	30.1	30.6	**30.9**
+DA+MSA$_{1:1}$	31.2	30.6	30.9	**31.3**
+DA+MSA$_{1:2}$	**31.3**	30.7	30.8	31.2
+DA+MSA$_{1:4}$	**31.2**	30.6	30.7	**31.2**

Table 3: Macro-average BLEU scores on the Dev sets. Bold denotes the best-performing model within each data mixture; underlining denotes the best-performing data mixture for each model.

In comparison, the LLMs substantially outperform the MSA-pretrained MT systems. Among the commercial models, Gemini slightly outperforms GPT-5.5, while for the Arabic-centric models, Jais outperforms Fanar. Although Jais achieves the highest average BLEU score overall, this is largely driven by its exceptionally strong performance on MADAR. A detailed breakdown of LLM performance across dialects and evaluation metrics is provided in Table 11 (Appendix E.2).

DA Fine-Tuning Table 3 reports Dev results after fine-tuning the MSA-pretrained models on DA data. All models benefit substantially from dialect-specific supervision, with average BLEU increasing by more than 10 points over MSA-only pretraining. The models remain closely matched across fine-tuning settings, with **Base** and **Pixel** achieving the highest scores. Detailed results in Table 12 (Appendix E.3) show similar trends across datasets and metrics. Consistent with the MSA pretraining results, DA supervision does not substantially differentiate the representation approaches, with all models exhibiting similar gains from fine-tuning.

Adding MSA data during fine-tuning further improves all models over DA-only training, although the gains are modest. The DA+MSA$_{1:1}$ mixture provides consistent improvements across representations, while increasing the proportion of MSA to 1:2 or 1:4 yields no statistically significant gains. This suggests that a balanced mixture is sufficient to capture the benefits of incorporating MSA during DA fine-tuning.

Test Results Table 4 presents Test results for the MSA-pretrained models and the DA+MSA$_{1:1}$-fine-tuned models. Among the MSA-pretrained models, **Base** achieves the highest average BLEU (19.5), followed by **Vocab+**, **Pixel**, and **Byte**. Unlike on the Dev sets, the differences between **Base** and the other three models are statistically significant. After DA+MSA$_{1:1}$ fine-tuning, the models remain closely matched, with **Base** achieving 31.1 BLEU and **Pixel** 30.9; their difference is not statistically significant, while their differences with **Vocab+/Byte** are. Among the LLMs, Gemini achieves the highest average BLEU, outperforming Jais. Detailed results across dialects and evaluation metrics are provided in Tables 13–15 (Appendix F).

5.2 RQ2: Data Scaling

Figure 2 shows how the models perform with varying amounts of MSA pretraining data, before and after DA+MSA$_{1:1}$ fine-tuning. Under MSA pretraining, **Base** consistently performs best with less data, with the largest advantage at 5%. As the amount of MSA data increases, the performance gaps among representations narrow, with all models achieving comparable performance on the full dataset. This suggests that vocabulary-free representations require more pretraining data to match standard subword tokenization.

DA+MSA$_{1:1}$ fine-tuning substantially improves performance across all data sizes while reducing differences among representations. With only 5%

Corpus	MSA Base	MSA Vocab+	MSA Byte	MSA Pixel	DA+MSA$_{1:1}$ Base	DA+MSA$_{1:1}$ Vocab+	DA+MSA$_{1:1}$ Byte	DA+MSA$_{1:1}$ Pixel	Gemini	Jais
Alexandria	**16.2**	16.1	15.7	15.9	**35.0**	34.4	34.5	34.8	43.0	41.8
APT	**9.1**	8.9	9.0	**9.1**	**20.4**	19.5	20.3	20.3	20.2	22.6
Bible	**10.6**	10.3	9.6	10.0	**12.5**	12.1	11.7	12.0	29.5	24.1
FLORES	**29.0**	28.8	28.4	28.8	31.3	30.4	30.7	**31.4**	41.0	39.5
MADAR	**23.4**	22.9	22.8	23.0	40.6	40.0	**40.8**	40.7	50.0	49.6
QAraC	**10.0**	9.5	**10.0**	**10.0**	**16.0**	15.1	15.7	15.5	20.0	19.0
Overall	**19.5**	19.3	18.9	19.2	**31.1**	30.4	30.6	30.9	40.2	38.8

Table 4: Macro-average BLEU scores on the Test sets. Bold indicates the best MSA-pretrained and DA+MSA$_{1:1}$-fine-tuned models; underline indicates the best LLM for each dataset.

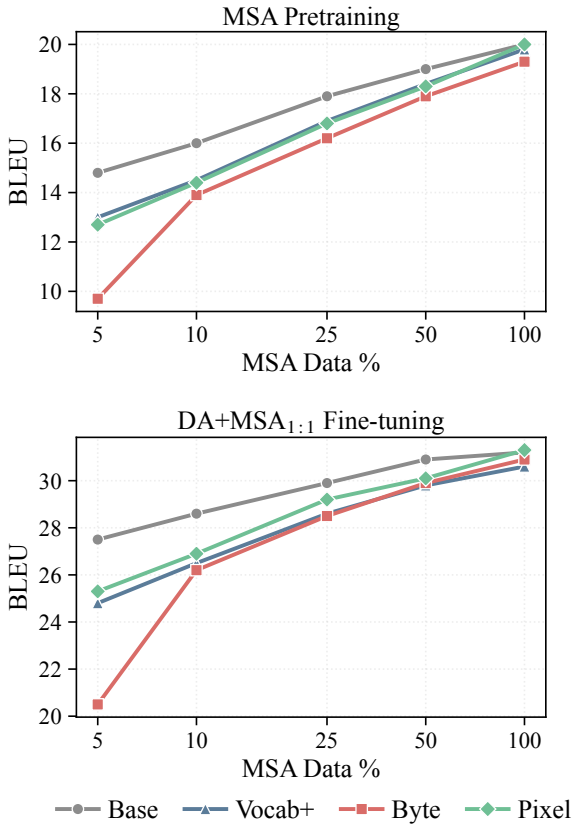


Figure 2: Effect of MSA pretraining data size on MT performance before and after DA+MSA$_{1:1}$ fine-tuning.

of the MSA data, **Vocab+** and **Pixel** already close much of the gap with **Base**, while **Byte** catches up with additional pretraining data. By 100%, all four models achieve comparable performance, indicating that dialect-specific supervision reduces the effect of limited MSA pretraining data on the choice of text representation. This highlights the importance of considering data availability when comparing text representations.

5.3 RQ3: Orthographic Robustness

Table 5 reports robustness results on the Test sets using the DA+MSA$_{1:1}$-fine-tuned models. All models degrade relative to the original inputs, but the magnitude of degradation varies substantially.

Among the models, **Pixel** is considerably more robust to both variations. On diacritized inputs, **Pixel** loses 7.5 BLEU on average, compared with 25.6–29.5 BLEU for the other representations, and achieves the highest BLEU on all six benchmarks. The advantage persists on dotless Arabic, where **Pixel** loses 17.6 BLEU compared with 27.8–28.9 for the other models. This aligns with prior work (Salesky et al., 2021), demonstrating the robustness of pixel-based representations to orthographic variation unseen during training. Such variations have less impact on pixel representations, which can recognize the base character form, unlike byte- and token-based representations that rely on exact character sequences.

LLMs are considerably more robust to diacritized Arabic, with Gemini and Jais losing only 1.8 and 2.9 BLEU on average, respectively. Their behavior differs sharply on dotless inputs, however: Gemini loses 7.1 BLEU, while Jais drops by 28.0 BLEU and falls below **Pixel**. Appendix G presents detailed robustness results across dialects.

5.4 RQ4: Computational Efficiency

As shown in Table 6, **Pixel** provides substantially higher training throughput than the other representation methods, while using the least memory at a comparable batch size (32). This efficiency stems from encoding each source word as a single embedding (§3.3), reducing the sequence length and associated decoder computation and memory. The lower memory requirements also allow a larger batch size (64), further increasing throughput.

Corpus	Base	Vocab+	Byte	Pixel	Gemini	Jais
Diacritized Input						
Alexandria	2.4 (↓32.6)	1.1 (↓33.3)	6.8 (↓27.7)	**26.9** (↓**7.9**)	41.9 (↓1.1)	39.3 (↓2.4)
APT	0.8 (↓19.6)	0.5 (↓19.0)	3.0 (↓17.2)	**15.1** (↓**5.2**)	8.4 (↓11.9)	18.8 (↓3.8)
Bible	0.7 (↓11.7)	0.6 (↓11.5)	1.8 (↓9.9)	**8.8** (↓**3.3**)	28.6 (↓0.9)	22.3 (↓1.9)
FLORES	0.9 (↓30.4)	0.5 (↓29.9)	2.2 (↓28.4)	**22.3** (↓**9.1**)	39.0 (↓2.0)	35.3 (↓4.3)
MADAR	2.7 (↓37.9)	1.2 (↓38.8)	8.4 (↓32.4)	**31.9** (↓**8.8**)	49.4 (↓0.5)	46.7 (↓2.9)
QAraC	0.7 (↓15.3)	0.3 (↓14.8)	3.2 (↓12.5)	**12.8** (↓**2.7**)	20.2 (↑0.2)	19.3 (↑0.3)
Overall	1.7 (↓29.3)	0.9 (↓29.5)	5.0 (↓25.6)	**23.4** (↓**7.5**)	38.4 (↓1.8)	35.8 (↓2.9)
Dotless Input						
Alexandria	4.5 (↓30.6)	2.1 (↓32.3)	3.5 (↓31.0)	**16.2** (↓**18.6**)	36.8 (↓6.2)	13.2 (↓28.5)
APT	2.1 (↓18.2)	0.9 (↓18.7)	1.8 (↓18.4)	**9.4** (↓**10.9**)	7.2 (↓13.0)	5.0 (↓17.6)
Bible	1.5 (↓11.0)	0.8 (↓11.3)	1.1 (↓10.6)	**5.2** (↓**6.8**)	27.2 (↓2.3)	9.3 (↓14.9)
FLORES	1.7 (↓29.6)	0.9 (↓29.5)	1.3 (↓29.4)	**12.2** (↓**19.1**)	31.7 (↓9.3)	8.0 (↓31.5)
MADAR	4.8 (↓35.8)	2.0 (↓38.1)	3.3 (↓37.5)	**16.4** (↓**24.3**)	42.5 (↓7.5)	13.6 (↓36.0)
QAraC	2.3 (↓13.8)	0.9 (↓14.1)	2.0 (↓13.7)	**7.6** (↓**8.0**)	15.5 (↓4.5)	3.5 (↓15.5)
Overall	3.3 (↓27.8)	1.5 (↓28.9)	2.5 (↓28.1)	**13.3** (↓**17.6**)	33.1 (↓7.1)	10.7 (↓28.0)

Table 5: Robustness results on the Test sets with diacritized and dotless Arabic inputs. Values in parentheses denote the BLEU change relative to the corresponding original input (↓: decrease; ↑: increase). Bold indicates the best-performing model; underline indicates the best-performing LLM.

Model	Batch	Sent./s ↑	Mem. ↓
Base	16	250.2±0.8	24.9
Vocab+	16	230.1±0.7	27.2
Byte	32	424.3±1.5	27.8
Pixel	32	666.2±5.9	**22.0**
Pixel	**64**	**726.8**±0.9	33.0

Table 6: Training efficiency. Sent./s denotes sentences per second (mean±standard deviation); Mem. denotes peak reserved GPU memory in GiB.

		Sent./s ↑		
Model	Batch	E2E	Prep.	Mem. ↓
Base	32	11.1±0.2	11,808±0.0	14.9
Vocab+	32	11.9±0.1	**12,800**±0.0	11.1
Byte	32	12.3±0.1	6,823±1.6	9.2
Pixel	32	12.4±0.1	1,529±0.2	**5.6**
Pixel	**64**	**15.6**±0.1	1,533±0.6	9.5

Table 7: Inference efficiency. Sent./s denotes sentences per second; E2E and Prep. denote end-to-end and preprocessing throughput (mean±standard deviation); Mem. denotes peak reserved GPU memory in GiB.

At inference (Table 7), despite the cost of online rendering, **Pixel** achieves the highest end-to-end throughput at a comparable batch size while using the least GPU memory. This allows **Pixel** to double the batch size to 64, further increasing throughput. Moreover, for deployment, all possible Arabic character bigrams (§3.3) can be pre-rendered and cached, eliminating online rendering overhead.

6 Conclusion and Future Work

We presented the first systematic comparison of vocabulary-based and vocabulary-free text representations for Arabic-English machine translation. Across 35 evaluation sets spanning MSA and 18 dialects, we find that the representations achieve broadly comparable translation performance with sufficient supervision. Standard subword representations perform best with limited MSA pretraining data, but differences diminish with more data and dialect-specific fine-tuning. Pixel-based representations are substantially more robust to unseen orthographic variation, including diacritized and dotless Arabic, while providing the highest training and end-to-end inference throughput. Our findings reveal trade-offs between translation performance, data requirements, robustness, and computational efficiency across Arabic text representations.

In future work, we plan to extend this comparison to other Arabic NLP tasks and investigate Arabic-specific pretraining strategies for pixel-based representations, particularly given their robustness and computational efficiency.

Limitations

Our study has several limitations. First, all experiments use a single decoder-only language model backbone (SmolLM2-360M-Instruct). Although this enables a controlled comparison, the relative performance of the different representation paradigms may differ for larger models or alternative architectures. Second, we evaluate text representations only on Arabic-English machine translation. While our benchmark covers MSA and 18 dialectal varieties, our findings may not generalize to other Arabic NLP tasks. Finally, our robustness evaluation is limited to two orthographic perturbations: diacritized and dotless Arabic. Future work should consider additional naturally occurring orthographic variations, such as spelling inconsistencies, informal writing conventions, and OCR noise.

Ethics Statement

We use all datasets and models in accordance with their respective licenses and terms of use. A summary of licenses is provided in Appendix C. While this work does not introduce new datasets containing personally identifiable information, the evaluated machine translation systems may produce inaccurate translations, particularly for Dialectal Arabic, and may reflect biases present in their training data. We therefore recommend appropriate human oversight when deploying these systems in applications where translation errors could have significant consequences. We used AI writing assistance within the scope of "Assistance purely with the language of the paper" described in the ACL Policy on Publication Ethics.

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A Models Parameters

Model	Params. (M)
Base	361.8
Vocab+	391.6
Byte	483.3
Pixel	485.5

Table 8: Number of parameters for each model.

B LLM Prompt

Prompt

```
You are an Arabic to English machine
translation system.

Translate the following
{Source Dialect} sentence to
English.

Output only the translation, nothing
else.

Here are some examples:
{Five in-context examples from the
benchmark training set, when
available}
```

C Licenses and Terms of Use

We list below the licenses of the data and models used in this work, all of which are employed in accordance with their intended use.

- Alexandria (Mekki et al., 2026): Creative Commons Attribution Non Commercial 4.0.
- APT (Zbib et al., 2012): LDC License (LDC2012T09).
- Bible: Publicly available parallel Arabic-English translations.
- FLORES (Goyal et al., 2022): Creative Commons Attribution Share Alike 4.0.
- MADAR (Bouamor et al., 2018): CMU and NYUAD Open-Source License.
- OPUS (Lison and Tiedemann, 2016): Collection of parallel corpora distributed under their respective original licenses.
- QAraC (Elmahdy et al., 2014): Creative Commons Attribution 4.0 International.
- QED (Guzman et al., 2013): Freely available for research use.
- TED (Cettolo, 2016): Creative Commons Attribution-NonCommercial-NoDerivatives.
- UN (Ziemski et al., 2016): Freely available for research use.
- SmolLM2-360M-Instruct (Allal et al., 2025): Apache 2.0.
- GPT-5.5 (Singh et al., 2026): Proprietary API model accessed via OpenAI API.
- Gemini-3.1-Flash (Team et al., 2025): Proprietary API model accessed via Google Gemini API.
- Jais-2-70B-Chat (Anwar et al., 2026): Apache 2.0.
- Fanar-2-27B-Instruct (TEAM et al., 2026): Apache 2.0.

D Data Statistics

Corpus	Dialect	Train	Dev	Test
Alexandria	Egyptian	3,108	1,113	1,118
Alexandria	Jordanian	5,501	1,113	1,107
Alexandria	Lebanese	8,906	1,118	1,106
Alexandria	Libyan	-	-	1,109
Alexandria	Mauritanian	2,573	1,110	1,115
Alexandria	Moroccan	5,515	1,114	1,112
Alexandria	Omani	6,280	1,109	1,118
Alexandria	Palestinian	14,933	1,110	1,109
Alexandria	Saudi	8,470	1,110	1,113
Alexandria	Sudanese	-	-	1,106
Alexandria	Syrian	6,071	1,119	1,114
Alexandria	Tunisian	2,034	1,116	1,109
Alexandria	Yemeni	3,089	1,118	1,106
APT	Egyptian	35,935	1,000	1,000
APT	Levantine	135,788	1,000	1,000
Bible	Moroccan	-	600	600
Bible	MSA	-	600	600
Bible	MSA	-	600	600
Bible	Tunisian	-	600	600
FLORES	Egyptian	-	997	1,012
FLORES	Iraqi	-	997	1,012
FLORES	Moroccan	-	997	1,012
FLORES	MSA	-	997	1,012
FLORES	N. Levantine	-	997	1,012
FLORES	Najdi	-	997	1,012
FLORES	S. Levantine	-	997	1,012
FLORES	Tunisian	-	997	1,012
FLORES	Yemeni	-	997	1,012
MADAR	Egyptian	9,000	1,000	2,000
MADAR	Lebanese	9,000	1,000	2,000
MADAR	Moroccan	9,000	1,000	2,000
MADAR	MSA	9,000	1,000	2,000
MADAR	Qatari	9,000	1,000	2,000
MADAR	Tunisian	9,000	1,000	2,000
OPUS	MSA	13,466,770	-	-
QAraC	Qatari	-	6,713	6,713
QED	MSA	376,337	-	-
TED	MSA	173,994	-	-
UN	MSA	10,024,940	-	-
Total		**24,334,244**	**38,336**	**46,663**

Table 9: Dataset statistics across corpora, dialects, and Train/Dev/Test splits. Bible contains two distinct MSA sets, following Sajjad et al. (2020).

E Dev Results

E.1 MSA-Pretrained Models

		BLEU				chrF++				COMET			
Corpus	**Dialect**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**
Alexandria	Egyptian	**18.9**	18.3	17.9	18.6	**39.9**	38.9	38.4	39.2	**0.72**	**0.72**	0.71	**0.72**
Alexandria	Jordanian	18.8	18.6	**19.2**	19.0	39.6	39.5	**39.9**	**39.9**	0.71	0.71	0.71	**0.72**
Alexandria	Lebanese	**15.2**	15.0	14.2	14.0	**34.9**	34.5	33.6	33.5	**0.67**	0.66	0.65	0.65
Alexandria	Mauritanian	**7.9**	7.6	6.5	7.3	**26.3**	26.1	23.6	24.7	**0.53**	**0.53**	0.51	0.52
Alexandria	Moroccan	9.2	**9.3**	9.2	8.9	**28.1**	27.8	27.5	27.6	**0.56**	**0.56**	0.55	**0.56**
Alexandria	Omani	18.8	**19.6**	19.4	19.4	41.0	**41.4**	41.3	**41.4**	0.71	**0.72**	0.71	0.71
Alexandria	Palestinian	17.0	17.3	17.3	**17.5**	37.8	37.8	37.6	**38.1**	**0.69**	**0.69**	**0.69**	**0.69**
Alexandria	Saudi	22.6	23.0	22.4	**23.1**	44.0	44.2	43.7	**44.7**	**0.75**	**0.75**	**0.75**	**0.75**
Alexandria	Syrian	**21.7**	20.9	20.6	21.6	**41.9**	41.2	41.0	**41.9**	**0.71**	0.70	0.70	**0.71**
Alexandria	Tunisian	**12.7**	12.6	11.8	11.6	**33.0**	32.6	31.4	31.1	**0.61**	**0.61**	0.59	0.59
Alexandria	Yemeni	18.2	18.7	17.9	**18.8**	39.8	39.8	39.1	**40.1**	**0.73**	**0.73**	0.72	**0.73**
APT	Egyptian	**11.1**	10.1	10.3	10.1	**32.8**	32.0	31.8	32.1	**0.65**	0.64	0.64	0.64
APT	Levantine	**9.0**	7.9	8.0	8.6	**30.2**	29.4	28.9	29.7	**0.62**	**0.62**	**0.62**	**0.62**
Bible	Moroccan	5.1	5.2	4.3	**5.3**	23.6	**23.7**	22.7	**23.7**	0.52	0.52	0.51	**0.54**
Bible	MSA	**17.0**	16.5	14.8	16.4	**38.4**	38.2	36.0	37.1	**0.74**	0.73	0.71	0.73
Bible	MSA	**12.8**	12.6	11.5	12.7	**33.5**	32.8	32.1	32.3	**0.68**	0.67	0.66	0.67
Bible	Tunisian	**7.6**	7.0	6.4	6.8	**26.6**	25.5	24.8	24.9	**0.56**	0.54	0.53	0.54
FLORES	Egyptian	25.2	25.0	24.8	**26.2**	51.0	50.5	50.1	**51.2**	0.79	0.79	0.79	**0.80**
FLORES	Iraqi	31.0	31.3	30.5	**31.7**	55.5	55.6	54.5	**55.7**	**0.82**	**0.82**	0.81	**0.82**
FLORES	Moroccan	17.9	17.7	17.3	**18.5**	42.3	41.8	41.0	**42.5**	**0.67**	**0.67**	0.66	**0.67**
FLORES	MSA	38.2	37.3	36.9	**38.3**	**61.2**	60.6	60.0	61.1	**0.86**	0.85	0.85	**0.86**
FLORES	N. Levantine	29.8	29.4	29.1	**30.2**	54.1	53.6	53.1	**54.2**	**0.80**	0.79	0.79	0.79
FLORES	Najdi	**37.3**	36.5	36.2	37.0	**60.6**	59.9	59.3	60.3	**0.85**	**0.85**	**0.85**	**0.85**
FLORES	S. Levantine	32.6	32.2	31.6	**33.3**	56.3	56.1	55.1	**56.8**	0.80	0.80	0.80	**0.81**
FLORES	Tunisian	**23.8**	23.1	23.6	**23.8**	**48.7**	48.1	47.9	48.6	**0.74**	0.73	0.73	**0.74**
FLORES	Yemeni	33.0	32.7	32.0	**33.4**	56.9	56.6	56.0	**57.5**	**0.83**	**0.83**	**0.83**	**0.83**
MADAR	Egyptian	**27.2**	26.2	25.5	26.0	**45.4**	44.1	43.9	44.2	**0.76**	0.74	0.74	0.74
MADAR	Lebanese	16.4	16.1	15.6	**16.5**	34.4	33.9	33.2	**34.6**	**0.64**	0.63	0.63	**0.64**
MADAR	Moroccan	12.4	**12.7**	11.3	12.0	30.5	**30.7**	29.4	30.5	**0.60**	**0.60**	0.58	0.59
MADAR	MSA	44.7	45.0	44.9	**45.1**	61.4	**61.7**	61.5	61.6	0.87	**0.88**	**0.88**	0.87
MADAR	Qatari	**26.9**	26.1	25.9	26.2	**44.4**	43.5	43.5	43.9	**0.75**	0.74	0.73	0.74
MADAR	Tunisian	11.6	11.4	**11.8**	**11.8**	**29.3**	29.1	29.0	29.1	**0.59**	0.58	0.58	0.58
QAraC	Qatari	**9.8**	9.4	9.6	9.3	**32.8**	32.0	32.5	32.1	**0.64**	0.63	0.63	0.63
Average		**20.0**	19.8	19.3	**20.0**	**41.1**	40.7	40.1	40.8	**0.70**	**0.70**	0.69	**0.70**

Table 10: MSA-pretrained models results on the Dev sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

E.2 LLMs

Corpus	Dialect	BLEU				chrF++				COMET			
		GPT-5.5	Gemini	Jais	Fanar	GPT-5.5	Gemini	Jais	Fanar	GPT-5.5	Gemini	Jais	Fanar
Alexandria	Egyptian	**43.0**	42.5	41.6	39.5	**62.0**	61.6	60.6	59.2	**0.86**	**0.86**	0.85	0.85
Alexandria	Jordanian	42.9	**44.1**	43.1	40.8	62.5	**62.9**	62.0	60.8	**0.87**	0.86	0.86	0.85
Alexandria	Lebanese	43.0	42.5	**44.1**	40.9	61.8	61.7	**62.0**	59.7	**0.86**	0.85	0.85	0.84
Alexandria	Mauritanian	40.8	**41.4**	41.0	37.1	60.0	**60.5**	60.1	56.8	**0.83**	**0.83**	0.82	0.80
Alexandria	Moroccan	34.4	**34.7**	29.4	23.6	54.5	**56.7**	49.8	44.8	0.79	**0.80**	0.75	0.73
Alexandria	Omani	**44.3**	43.1	43.1	39.4	**63.8**	62.9	62.6	60.1	**0.87**	0.86	0.85	0.85
Alexandria	Palestinian	42.4	**42.6**	42.4	40.5	**62.2**	62.1	61.5	60.2	**0.86**	**0.86**	0.85	0.85
Alexandria	Saudi	46.2	**47.6**	46.0	43.6	65.2	**65.8**	65.0	63.1	**0.87**	**0.87**	0.86	0.86
Alexandria	Syrian	53.8	**54.0**	52.3	49.4	**70.0**	69.8	68.7	67.0	**0.88**	0.87	0.87	0.86
Alexandria	Tunisian	44.3	**44.7**	42.3	39.6	**62.2**	62.1	60.4	58.4	**0.85**	**0.85**	0.83	0.83
Alexandria	Yemeni	**39.2**	38.5	37.5	35.5	**59.5**	58.9	58.5	56.4	**0.86**	0.85	0.85	0.84
APT	Egyptian	**24.2**	21.6	20.4	18.6	**50.1**	49.4	47.4	45.6	**0.77**	**0.77**	0.76	0.75
APT	Levantine	16.6	**18.3**	14.2	15.2	45.0	**47.0**	45.7	43.9	0.76	**0.77**	0.75	0.74
Bible	Moroccan	24.2	**29.4**	23.5	22.3	47.2	**50.3**	44.7	43.5	**0.77**	**0.77**	0.75	0.74
Bible	MSA	27.1	**33.5**	27.4	31.1	49.3	**53.1**	48.4	50.9	**0.81**	**0.81**	0.80	**0.81**
Bible	MSA	23.4	28.1	25.1	**28.3**	46.0	**48.3**	46.1	48.2	**0.78**	**0.78**	0.77	**0.78**
Bible	Tunisian	24.8	**29.4**	25.3	23.3	46.9	**49.6**	45.5	43.7	**0.77**	**0.77**	0.75	0.74
FLORES	Egyptian	37.2	37.3	**39.8**	33.8	62.3	62.8	**63.2**	59.8	**0.87**	**0.87**	0.86	0.85
FLORES	Iraqi	41.3	40.6	**42.7**	37.7	65.7	**65.9**	65.8	62.8	**0.88**	**0.88**	0.87	0.87
FLORES	Moroccan	36.4	35.6	**39.1**	30.9	60.3	60.5	**62.0**	56.3	**0.84**	**0.84**	**0.84**	0.82
FLORES	MSA	46.0	46.7	**47.4**	44.5	69.1	**69.8**	69.3	67.2	**0.89**	**0.89**	**0.89**	**0.89**
FLORES	N. Levantine	45.6	46.5	**47.7**	41.1	67.7	**68.9**	67.9	64.4	**0.88**	**0.88**	0.87	0.86
FLORES	Najdi	45.0	45.6	**46.0**	42.7	68.2	**69.1**	68.6	66.5	**0.89**	**0.89**	0.88	0.88
FLORES	S. Levantine	48.8	49.6	**51.7**	44.2	70.0	**71.1**	70.4	66.7	0.89	**0.89**	0.88	0.87
FLORES	Tunisian	38.2	34.7	**38.6**	32.6	**62.7**	62.0	62.1	58.5	**0.86**	0.85	0.85	0.83
FLORES	Yemeni	42.7	41.7	**44.1**	38.8	66.6	66.8	**66.9**	63.6	**0.88**	**0.88**	0.87	0.87
MADAR	Egyptian	52.8	52.7	**78.3**	45.7	68.9	69.3	**85.3**	63.3	0.89	0.89	**0.92**	0.87
MADAR	Lebanese	52.6	52.2	**71.8**	40.8	68.4	68.9	**81.3**	58.8	0.88	0.88	**0.91**	0.83
MADAR	Moroccan	48.9	49.0	**70.2**	38.7	65.6	65.7	**79.7**	56.9	0.87	0.87	**0.90**	0.82
MADAR	MSA	57.6	56.2	**75.3**	53.5	73.2	72.5	**84.5**	69.7	0.92	0.91	**0.94**	0.91
MADAR	Qatari	54.3	55.6	**68.6**	46.8	69.6	70.5	**78.7**	63.3	0.90	0.90	**0.91**	0.87
MADAR	Tunisian	46.0	45.5	**66.2**	32.9	63.2	63.3	**76.4**	51.2	0.85	0.85	**0.88**	0.78
QAraC	Qatari	17.9	**19.3**	18.9	15.7	46.8	**48.9**	47.0	43.5	0.76	**0.77**	0.75	0.73
Average		40.2	40.8	**43.8**	36.0	61.1	61.8	**63.0**	57.4	**0.85**	**0.85**	0.84	0.83

Table 11: LLM results on the Dev sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

E.3 DA-Fine-Tuned Models

Data Mixture	Corpus	BLEU				chrF++				COMET			
		Base	Vocab+	Byte	Pixel	Base	Vocab+	Byte	Pixel	Base	Vocab+	Byte	Pixel
MSA	Alexandria	**16.4**	**16.4**	16.0	16.3	**36.9**	36.7	36.1	36.5	**0.67**	**0.67**	0.66	**0.67**
	APT	**10.0**	9.0	9.1	9.3	**31.5**	30.7	30.4	30.9	**0.63**	**0.63**	**0.63**	**0.63**
	Bible	**10.6**	10.3	9.2	10.3	**30.5**	30.1	28.9	29.5	**0.63**	0.62	0.60	0.62
	FLORES	29.9	29.5	29.1	**30.2**	54.0	53.6	53.0	**54.2**	0.79	0.79	0.79	**0.80**
	MADAR	**23.2**	22.9	22.5	22.9	**40.9**	40.5	40.1	40.6	**0.70**	0.69	0.69	0.69
	QAraC	**9.8**	9.4	9.6	9.3	**32.8**	32.0	32.5	32.1	**0.64**	0.63	0.63	0.63
	Overall	**20.0**	19.8	19.3	**20.0**	**41.1**	40.7	40.1	40.8	**0.70**	**0.70**	0.69	**0.70**
+DA	Alexandria	**35.7**	34.7	35.0	35.5	**54.7**	53.8	54.0	54.4	**0.80**	0.79	0.79	**0.80**
	APT	**20.8**	19.9	20.3	20.3	**45.7**	44.6	45.2	45.2	**0.73**	0.72	**0.73**	**0.73**
	Bible	12.0	11.4	11.5	11.6	33.4	32.7	32.9	33.5	0.66	0.65	0.65	0.66
	FLORES	30.1	29.9	30.4	30.6	55.5	55.0	55.2	55.9	**0.82**	0.81	**0.82**	**0.82**
	MADAR	41.4	40.3	41.3	**41.8**	58.6	57.8	58.4	**58.8**	**0.82**	0.81	**0.82**	**0.82**
	QAraC	**16.9**	15.6	16.4	16.2	**41.2**	40.1	40.9	40.5	**0.69**	0.68	**0.69**	**0.69**
	Overall	30.9	30.1	30.6	30.9	52.1	51.3	51.7	52.1	0.78	**0.78**	**0.78**	**0.78**
+DA+MSA$_{1:1}$	Alexandria	35.5	34.8	34.9	35.4	54.5	53.9	53.8	54.2	**0.80**	0.79	0.79	**0.80**
	APT	20.2	19.5	20.1	19.9	45.2	44.0	44.7	44.7	**0.73**	0.72	0.72	**0.73**
	Bible	12.2	12.2	11.7	12.3	33.6	33.2	33.1	33.9	**0.67**	0.66	0.66	**0.67**
	FLORES	31.9	31.3	31.6	32.1	56.3	55.6	55.7	56.5	**0.82**	**0.82**	**0.82**	**0.82**
	MADAR	41.3	40.6	41.5	41.7	58.6	57.8	58.4	58.4	**0.82**	**0.82**	**0.82**	**0.82**
	QAraC	15.6	14.8	15.4	15.2	40.3	39.3	39.9	39.4	**0.69**	0.68	**0.69**	0.68
	Overall	31.2	30.6	30.9	**31.3**	52.2	51.5	51.7	52.1	**0.79**	0.78	0.78	0.78
+DA+MSA$_{1:2}$	Alexandria	35.5	34.9	34.9	35.2	54.5	53.9	53.9	54.1	**0.80**	0.79	0.79	**0.80**
	APT	20.2	19.4	19.9	20.1	44.9	44.1	44.5	44.6	**0.73**	0.72	0.72	**0.73**
	Bible	**12.4**	12.3	11.7	12.1	33.7	33.3	33.1	33.8	**0.67**	0.66	0.66	**0.67**
	FLORES	32.1	31.4	31.6	**32.2**	56.4	55.8	55.9	**56.6**	**0.82**	**0.82**	**0.82**	**0.82**
	MADAR	41.4	40.6	41.1	41.6	**58.7**	57.8	58.2	58.4	**0.82**	**0.82**	**0.82**	**0.82**
	QAraC	15.3	14.7	15.1	15.0	40.0	39.3	39.7	39.2	**0.69**	0.68	**0.69**	0.68
	Overall	**31.3**	30.7	30.8	31.2	**52.3**	51.6	51.7	52.1	**0.79**	0.78	0.78	0.78
+DA+MSA$_{1:4}$	Alexandria	35.2	34.9	34.8	35.2	54.3	53.9	53.7	54.1	**0.80**	0.79	0.79	**0.80**
	APT	20.2	19.3	19.5	20.0	44.9	43.9	44.2	44.6	**0.73**	0.72	0.72	**0.73**
	Bible	12.3	12.1	11.7	12.2	33.7	33.0	33.1	33.8	**0.67**	0.66	0.66	**0.67**
	FLORES	**32.2**	31.5	31.6	32.1	**56.5**	55.8	55.7	56.4	**0.82**	**0.82**	**0.82**	**0.82**
	MADAR	41.2	40.5	41.1	41.3	58.4	57.6	58.1	58.3	**0.82**	0.81	**0.82**	**0.82**
	QAraC	14.9	14.3	14.9	14.8	39.6	38.8	39.5	38.8	0.68	0.67	0.68	0.68
	Overall	31.2	30.6	30.7	31.1	52.1	51.5	51.6	52.0	**0.79**	0.78	0.78	0.78

Table 12: Macro-average BLEU, chrF++, and COMET scores on the Dev sets. Bold denotes the best-performing model within each data mixture; underlining denotes the best-performing fine-tuning data mixture for each model.

F Test Results

F.1 MSA-Pretrained Models

		BLEU				chrF++				COMET			
Corpus	**Dialect**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**
Alexandria	Egyptian	**17.0**	16.4	16.2	16.4	**37.7**	37.0	36.8	37.0	**0.70**	**0.70**	0.69	**0.70**
Alexandria	Jordanian	**18.8**	17.7	18.3	18.5	**40.0**	39.2	39.5	39.9	**0.72**	0.71	0.71	**0.72**
Alexandria	Lebanese	**15.9**	15.6	15.1	14.9	**35.2**	34.9	33.6	33.8	**0.66**	0.65	0.64	0.64
Alexandria	Libyan	**14.7**	14.3	14.0	14.3	**35.7**	35.1	35.0	35.3	0.68	0.68	0.68	**0.69**
Alexandria	Mauritanian	**7.1**	**7.1**	6.6	6.9	**25.1**	**25.1**	23.1	24.2	**0.52**	**0.52**	0.51	0.51
Alexandria	Moroccan	**8.8**	**8.8**	8.5	8.5	**27.7**	27.3	26.9	27.2	**0.55**	**0.55**	**0.55**	**0.55**
Alexandria	Omani	**18.6**	18.2	18.2	18.1	**40.0**	39.6	39.8	39.7	**0.71**	0.70	0.70	0.70
Alexandria	Palestinian	16.7	17.0	**17.1**	16.9	37.6	37.6	**37.8**	37.6	**0.68**	**0.68**	**0.68**	**0.68**
Alexandria	Saudi	22.4	**23.2**	21.8	22.2	43.5	**43.9**	42.9	43.2	**0.74**	**0.74**	**0.74**	**0.74**
Alexandria	Sudanese	19.1	19.1	18.5	**19.4**	39.9	39.8	39.4	**40.3**	**0.71**	**0.71**	0.70	**0.71**
Alexandria	Syrian	20.4	20.1	20.0	**21.1**	40.7	40.1	40.2	**41.2**	0.70	0.70	0.70	**0.71**
Alexandria	Tunisian	**14.3**	**14.3**	12.3	12.0	**34.5**	**34.5**	31.6	31.2	**0.61**	**0.61**	0.58	0.58
Alexandria	Yemeni	**17.3**	16.8	17.1	17.2	37.9	37.3	37.3	**38.0**	**0.71**	**0.71**	0.70	**0.71**
APT	Egyptian	**10.5**	10.1	10.1	10.1	**33.3**	32.6	32.5	32.4	**0.64**	**0.64**	0.63	**0.64**
APT	Levantine	7.7	7.7	7.8	**8.1**	30.1	29.9	29.6	**30.3**	0.62	0.62	0.62	**0.63**
Bible	Moroccan	4.5	**4.8**	4.4	**4.8**	**23.2**	23.0	22.3	22.9	0.51	0.51	0.51	**0.53**
Bible	MSA	**17.1**	16.7	15.5	16.2	**38.1**	37.8	36.1	36.4	**0.74**	0.72	0.71	0.72
Bible	MSA	**12.9**	12.4	11.7	11.8	**33.2**	32.2	31.9	31.5	**0.67**	0.66	0.65	0.66
Bible	Tunisian	**8.0**	7.1	6.8	7.2	**27.3**	26.0	25.3	25.6	**0.56**	0.55	0.54	0.55
FLORES	Egyptian	**25.1**	**25.1**	24.7	24.9	**50.9**	50.3	49.6	50.2	**0.80**	0.79	0.79	0.79
FLORES	Iraqi	29.9	**30.0**	29.7	29.4	**54.9**	54.7	54.3	54.3	**0.82**	**0.82**	0.81	**0.82**
FLORES	Moroccan	**18.2**	17.7	17.3	17.6	**42.6**	41.8	41.1	41.8	0.67	0.67	0.67	**0.68**
FLORES	MSA	36.5	**36.8**	35.8	36.4	**60.5**	60.4	59.5	59.9	**0.86**	**0.86**	0.85	0.85
FLORES	N. Levantine	**28.5**	27.3	28.2	28.3	**53.3**	52.3	52.6	53.1	0.79	0.79	0.79	**0.80**
FLORES	Najdi	35.9	**36.2**	35.0	35.9	**60.0**	59.8	58.8	59.5	**0.85**	**0.85**	**0.85**	**0.85**
FLORES	S. Levantine	**31.9**	31.3	30.5	31.8	**55.6**	55.2	54.1	55.2	**0.80**	**0.80**	0.79	**0.80**
FLORES	Tunisian	22.9	22.9	22.7	**23.2**	47.9	**48.0**	47.4	**48.0**	**0.74**	**0.74**	**0.74**	**0.74**
FLORES	Yemeni	32.0	**32.3**	31.4	32.0	56.5	**56.7**	55.8	56.4	**0.83**	**0.83**	**0.83**	**0.83**
MADAR	Egyptian	**26.4**	25.7	25.6	25.9	**44.6**	43.9	43.7	44.3	**0.74**	**0.74**	**0.74**	**0.74**
MADAR	Lebanese	**16.8**	15.8	16.0	16.6	**34.4**	33.6	33.2	34.1	**0.64**	0.63	0.62	0.63
MADAR	Moroccan	**13.1**	12.1	12.4	12.4	**31.5**	30.4	30.4	30.8	**0.61**	0.59	0.59	0.59
MADAR	MSA	**45.4**	**45.4**	45.2	45.3	**62.0**	61.8	61.7	61.6	**0.88**	0.87	0.87	0.87
MADAR	Qatari	26.0	**26.2**	25.7	25.5	**44.1**	43.9	43.3	43.5	**0.74**	**0.74**	0.73	0.73
MADAR	Tunisian	**12.5**	11.9	11.9	12.1	**30.0**	29.4	29.4	29.3	**0.58**	**0.58**	**0.58**	0.57
QAraC	Qatari	**10.0**	9.5	**10.0**	**10.0**	**33.1**	32.3	32.9	32.7	**0.64**	0.63	**0.64**	0.63
Average		**19.5**	19.3	18.9	19.2	**40.6**	40.2	39.7	40.1	**0.70**	0.69	0.69	0.69

Table 13: MSA-pretrained models results on the Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

F.2 DA-Fine-Tuned Models

		BLEU				chrF++				COMET			
Corpus	**Dialect**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**
Alexandria	Egyptian	**34.1**	33.5	33.8	**34.1**	**54.5**	53.7	54.0	54.1	**0.81**	**0.81**	**0.81**	**0.81**
Alexandria	Jordanian	38.7	**38.8**	38.7	38.7	57.6	**57.7**	57.3	**57.7**	**0.83**	**0.83**	**0.83**	**0.83**
Alexandria	Lebanese	**39.0**	37.7	38.3	38.4	**56.9**	55.8	56.0	56.0	**0.82**	0.81	0.81	0.81
Alexandria	Libyan	28.1	27.9	28.4	**29.1**	48.9	48.4	48.9	**49.3**	**0.79**	**0.79**	**0.79**	**0.79**
Alexandria	Mauritanian	**24.3**	23.6	23.2	23.4	**43.7**	42.8	42.2	42.6	**0.70**	0.69	0.68	0.69
Alexandria	Moroccan	**23.7**	22.8	23.1	23.5	**43.5**	42.4	42.8	43.3	**0.69**	**0.69**	**0.69**	**0.69**
Alexandria	Omani	**35.4**	34.9	35.3	35.2	**55.1**	54.7	55.0	**55.1**	**0.81**	**0.81**	**0.81**	**0.81**
Alexandria	Palestinian	38.4	37.9	37.9	**38.9**	57.1	56.2	56.5	**57.2**	**0.82**	**0.82**	**0.82**	**0.82**
Alexandria	Saudi	**40.8**	39.7	40.7	40.7	59.2	58.6	59.1	**59.3**	**0.84**	0.83	0.83	0.83
Alexandria	Sudanese	**37.5**	36.7	36.6	37.4	**56.5**	55.7	55.9	56.4	**0.81**	0.80	**0.81**	**0.81**
Alexandria	Syrian	46.0	45.8	**46.3**	46.2	63.0	62.5	62.9	**63.1**	**0.85**	0.84	0.84	**0.85**
Alexandria	Tunisian	**36.7**	35.6	33.8	34.2	**56.4**	55.6	53.6	54.1	**0.78**	0.77	0.76	0.77
Alexandria	Yemeni	**32.7**	32.4	32.1	**32.7**	**52.0**	51.4	51.5	**52.0**	**0.81**	0.80	0.80	0.80
APT	Egyptian	20.8	20.3	20.7	**21.0**	**45.8**	45.1	45.5	45.5	**0.73**	0.72	0.72	0.72
APT	Levantine	**20.0**	18.8	19.8	19.6	**45.5**	44.1	45.2	44.9	**0.73**	0.72	**0.73**	**0.73**
Bible	Moroccan	8.4	7.6	8.1	**8.6**	29.6	28.3	28.9	**29.7**	**0.60**	0.58	0.59	**0.60**
Bible	MSA	**17.3**	17.2	16.6	16.6	**39.0**	38.5	38.3	38.4	**0.75**	0.74	0.74	0.74
Bible	MSA	12.8	**13.1**	12.3	12.4	**33.2**	33.1	33.0	33.1	**0.67**	**0.67**	0.66	**0.67**
Bible	Tunisian	**11.3**	10.4	9.8	10.4	**32.0**	30.8	30.7	31.4	**0.63**	0.62	0.61	**0.63**
FLORES	Egyptian	27.3	26.8	27.6	**28.0**	53.4	52.7	53.0	**53.6**	**0.82**	**0.82**	**0.82**	**0.82**
FLORES	Iraqi	31.1	30.2	30.1	**31.2**	56.1	55.3	55.4	**56.3**	**0.83**	**0.83**	**0.83**	**0.83**
FLORES	Moroccan	**22.2**	20.9	20.8	21.7	**47.0**	45.8	45.9	46.7	**0.74**	**0.74**	**0.74**	**0.74**
FLORES	MSA	**36.6**	36.1	36.3	36.5	60.5	59.9	60.1	**60.6**	**0.86**	0.85	0.85	**0.86**
FLORES	N. Levantine	**33.2**	31.5	32.2	33.1	**57.6**	56.0	56.7	57.5	**0.83**	0.82	**0.83**	**0.83**
FLORES	Najdi	35.9	35.3	35.7	**36.2**	59.9	59.3	59.7	**60.4**	**0.85**	**0.85**	**0.85**	**0.85**
FLORES	S. Levantine	**36.6**	34.8	35.5	36.1	**59.8**	58.2	58.8	59.5	0.83	0.83	0.83	**0.84**
FLORES	Tunisian	**26.3**	25.5	25.6	26.1	**51.5**	50.9	51.1	**51.5**	**0.79**	0.78	**0.79**	**0.79**
FLORES	Yemeni	33.0	32.8	32.3	**33.2**	57.7	57.3	57.2	**58.2**	**0.84**	**0.84**	**0.84**	**0.84**
MADAR	Egyptian	**43.1**	42.2	43.0	43.0	**61.1**	60.1	60.8	60.7	**0.85**	0.84	0.84	**0.85**
MADAR	Lebanese	**41.3**	40.2	41.1	**41.3**	**58.6**	57.1	58.0	58.2	**0.82**	0.81	0.81	**0.82**
MADAR	Moroccan	**33.9**	32.4	33.8	**33.9**	**52.7**	51.3	52.4	52.6	**0.78**	0.77	0.77	**0.78**
MADAR	MSA	50.2	50.9	**51.2**	50.6	66.8	**67.3**	**67.3**	67.0	**0.89**	**0.89**	**0.89**	**0.89**
MADAR	Qatari	43.0	**43.6**	**43.6**	43.4	59.9	**60.3**	**60.3**	60.1	**0.84**	**0.84**	**0.84**	**0.84**
MADAR	Tunisian	31.9	31.1	**32.3**	31.7	49.5	48.9	**50.2**	49.4	**0.76**	0.75	**0.76**	0.75
QAraC	Qatari	**16.0**	15.1	15.7	15.5	**40.4**	39.6	40.2	39.8	**0.69**	0.68	**0.69**	0.68
Average		**31.1**	30.4	30.6	30.9	**52.1**	51.3	51.6	51.9	**0.78**	**0.78**	**0.78**	**0.78**

Table 14: The DA+MSA$_{1:1}$ fine-tuned models result on the Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

F.3 LLMs

		BLEU		chrF++		COMET	
Corpus	**Dialect**	**Gemini**	**Jais**	**Gemini**	**Jais**	**Gemini**	**Jais**
Alexandria	Egyptian	**40.7**	40.1	**61.9**	60.5	**0.86**	0.85
Alexandria	Jordanian	**43.6**	41.8	**62.6**	61.3	**0.86**	**0.86**
Alexandria	Lebanese	45.3	**46.8**	**64.0**	63.5	**0.86**	0.85
Alexandria	Libyan	**32.3**	32.1	**53.6**	53.2	**0.83**	**0.83**
Alexandria	Mauritanian	**41.4**	40.1	**60.6**	59.1	**0.83**	0.81
Alexandria	Moroccan	**38.5**	30.1	**57.9**	50.5	**0.80**	0.75
Alexandria	Omani	**41.4**	41.1	**61.4**	60.6	**0.85**	**0.85**
Alexandria	Palestinian	43.4	**43.7**	**63.4**	62.5	**0.86**	0.85
Alexandria	Saudi	**47.1**	45.2	**65.7**	64.1	**0.87**	0.86
Alexandria	Sudanese	42.4	**42.5**	**63.4**	62.0	**0.86**	**0.86**
Alexandria	Syrian	**53.0**	52.0	**69.5**	68.3	**0.88**	0.87
Alexandria	Tunisian	**51.6**	49.3	**69.9**	67.3	**0.86**	0.85
Alexandria	Yemeni	38.1	**38.1**	**58.5**	58.2	**0.85**	0.84
APT	Egyptian	22.1	**26.6**	50.0	**53.0**	**0.77**	**0.77**
APT	Levantine	18.3	**18.5**	**47.3**	46.8	**0.76**	0.75
Bible	Moroccan	**29.4**	24.0	**50.3**	45.2	**0.76**	0.74
Bible	MSA	**33.7**	27.9	**53.2**	48.5	**0.81**	0.79
Bible	MSA	**26.7**	23.8	**47.8**	44.9	**0.77**	0.76
Bible	Tunisian	**28.1**	20.7	**49.2**	43.1	**0.76**	0.73
FLORES	Egyptian	35.6	**35.8**	**61.9**	60.2	**0.86**	**0.86**
FLORES	Iraqi	**38.9**	38.5	**65.0**	62.8	**0.87**	**0.87**
FLORES	Moroccan	**35.8**	33.9	**60.7**	58.2	**0.84**	0.83
FLORES	MSA	**46.4**	44.0	**69.7**	66.9	**0.89**	0.88
FLORES	N. Levantine	**44.0**	41.3	**67.6**	63.8	**0.88**	0.86
FLORES	Najdi	**44.3**	42.7	**68.6**	66.4	**0.89**	0.88
FLORES	S. Levantine	**47.2**	45.1	**69.7**	66.5	**0.88**	0.87
FLORES	Tunisian	**36.2**	34.9	**62.2**	59.6	**0.85**	0.84
FLORES	Yemeni	**40.9**	39.6	**66.4**	64.0	**0.88**	0.87
MADAR	Egyptian	49.6	**50.3**	**67.1**	66.7	**0.89**	0.88
MADAR	Lebanese	48.7	**50.9**	66.0	**66.9**	**0.88**	0.87
MADAR	Moroccan	**48.5**	46.2	**65.5**	63.2	**0.87**	0.85
MADAR	MSA	**54.6**	54.0	**71.5**	70.2	**0.91**	**0.91**
MADAR	Qatari	**52.9**	50.8	**68.5**	67.1	**0.89**	0.88
MADAR	Tunisian	**45.5**	45.3	**63.2**	62.2	**0.86**	0.84
QAraC	Qatari	**20.0**	19.0	**49.4**	47.1	**0.77**	0.75
Average		**40.2**	38.8	**61.5**	59.5	**0.85**	0.83

Table 15: Gemini and Jais results on the Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

G Robustness Results

G.1 Diacritized Input

		BLEU				chrF++				COMET			
Corpus	**Dialect**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**	**Base**	**Vocab+**	**Byte**	**Pixel**
Alexandria	Egyptian	1.5	0.7	4.5	**26.2**	16.3	14.7	25.3	**46.3**	0.45	0.39	0.54	**0.72**
Alexandria	Jordanian	3.0	1.2	7.7	**30.6**	16.6	15.1	28.2	**50.1**	0.46	0.40	0.58	**0.75**
Alexandria	Lebanese	2.8	1.3	8.1	**29.2**	17.8	16.4	28.7	**48.0**	0.47	0.42	0.57	**0.72**
Alexandria	Libyan	2.1	0.8	6.3	**23.4**	15.9	14.9	26.8	**43.4**	0.45	0.40	0.57	**0.72**
Alexandria	Mauritanian	4.1	2.0	6.8	**18.2**	20.1	18.2	26.5	**37.5**	0.49	0.44	0.55	**0.64**
Alexandria	Moroccan	2.9	1.7	5.4	**17.3**	17.5	16.6	25.4	**36.9**	0.46	0.41	0.54	**0.64**
Alexandria	Omani	1.8	1.0	6.1	**27.7**	15.5	14.6	26.7	**47.9**	0.45	0.40	0.57	**0.74**
Alexandria	Palestinian	2.3	0.9	6.9	**29.5**	16.1	14.8	27.4	**49.0**	0.45	0.39	0.56	**0.74**
Alexandria	Saudi	2.0	0.6	7.1	**30.9**	15.3	14.3	27.4	**50.2**	0.44	0.39	0.57	**0.75**
Alexandria	Sudanese	1.5	0.7	5.9	**29.4**	15.0	14.2	26.4	**48.8**	0.43	0.38	0.55	**0.73**
Alexandria	Syrian	2.3	0.9	9.1	**35.4**	16.9	15.6	29.8	**53.9**	0.45	0.40	0.58	**0.76**
Alexandria	Tunisian	3.0	1.6	7.0	**26.3**	19.0	17.3	28.2	**46.5**	0.48	0.42	0.56	**0.70**
Alexandria	Yemeni	2.2	1.1	7.1	**25.9**	16.6	15.3	27.4	**46.0**	0.45	0.40	0.57	**0.73**
APT	Egyptian	0.8	0.5	3.0	**15.5**	13.0	13.9	23.2	**38.6**	0.40	0.39	0.51	**0.65**
APT	Levantine	0.9	0.6	3.0	**14.6**	13.8	14.5	24.3	**38.9**	0.40	0.40	0.53	**0.66**
Bible	Moroccan	0.6	0.4	1.7	**6.3**	13.2	15.4	20.3	**25.9**	0.36	0.36	0.46	**0.55**
Bible	MSA	0.7	0.5	1.9	**12.6**	10.6	13.1	20.2	**32.6**	0.35	0.35	0.47	**0.64**
Bible	MSA	0.6	0.6	1.4	**8.5**	11.0	14.2	19.4	**27.8**	0.35	0.36	0.46	**0.59**
Bible	Tunisian	1.1	0.8	2.2	**7.7**	13.2	14.7	21.1	**27.2**	0.38	0.37	0.48	**0.56**
FLORES	Egyptian	0.9	0.3	2.2	**19.8**	13.1	12.4	22.8	**45.3**	0.37	0.33	0.49	**0.72**
FLORES	Iraqi	0.7	0.7	1.9	**22.0**	13.4	13.3	22.6	**47.5**	0.37	0.34	0.49	**0.74**
FLORES	Moroccan	1.2	0.6	2.2	**15.7**	15.5	14.3	22.7	**40.1**	0.40	0.35	0.49	**0.67**
FLORES	MSA	0.8	0.4	2.0	**26.0**	12.2	12.8	22.4	**51.0**	0.36	0.34	0.49	**0.75**
FLORES	N. Levantine	1.0	0.5	2.4	**23.3**	13.5	12.9	22.9	**47.5**	0.37	0.34	0.49	**0.73**
FLORES	Najdi	0.6	0.5	2.4	**25.8**	13.5	13.0	23.0	**50.9**	0.37	0.34	0.49	**0.75**
FLORES	S. Levantine	1.2	0.6	2.8	**25.8**	13.8	13.4	23.4	**49.8**	0.38	0.34	0.49	**0.73**
FLORES	Tunisian	0.9	0.5	2.0	**19.0**	14.1	13.6	22.6	**43.9**	0.38	0.34	0.48	**0.70**
FLORES	Yemeni	0.9	0.5	2.2	**23.3**	14.0	13.0	22.9	**49.1**	0.38	0.34	0.49	**0.74**
MADAR	Egyptian	2.0	1.0	8.3	**35.8**	16.6	15.0	29.3	**54.2**	0.47	0.43	0.58	**0.77**
MADAR	Lebanese	2.6	1.0	9.5	**32.3**	18.1	16.0	29.1	**49.8**	0.49	0.43	0.59	**0.74**
MADAR	Moroccan	3.8	2.0	8.8	**25.0**	20.0	17.8	29.2	**44.4**	0.51	0.45	0.58	**0.70**
MADAR	MSA	1.7	0.8	6.1	**41.5**	15.1	13.7	26.9	**59.8**	0.47	0.41	0.57	**0.81**
MADAR	Qatari	2.8	1.1	9.9	**33.1**	17.4	15.3	29.6	**51.0**	0.49	0.43	0.60	**0.76**
MADAR	Tunisian	3.0	1.5	8.0	**23.5**	18.3	16.2	27.6	**41.2**	0.50	0.44	0.58	**0.69**
QAraC	Qatari	0.7	0.3	3.2	**12.8**	12.7	12.8	22.0	**34.6**	0.42	0.40	0.52	**0.62**
Average		1.7	0.9	5.0	**23.4**	15.3	14.7	25.2	**44.4**	0.43	0.39	0.53	**0.70**

Table 16: The DA+MSA$_{1:1}$ fine-tuned models results on the diacritized Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

		BLEU		chrF++		COMET	
Corpus	**Dialect**	**Gemini**	**Jais**	**Gemini**	**Jais**	**Gemini**	**Jais**
Alexandria	Egyptian	**40.1**	37.9	**61.8**	59.2	**0.82**	0.81
Alexandria	Jordanian	**42.7**	39.2	**61.9**	59.4	**0.82**	0.81
Alexandria	Lebanese	**44.6**	44.0	**64.0**	62.4	**0.82**	0.81
Alexandria	Libyan	**31.5**	30.1	**53.4**	51.5	**0.80**	0.78
Alexandria	Mauritanian	**40.3**	36.2	**59.8**	56.1	**0.80**	0.79
Alexandria	Moroccan	**36.1**	25.5	**55.3**	46.6	**0.77**	0.71
Alexandria	Omani	**40.2**	37.4	**61.1**	58.4	**0.82**	0.81
Alexandria	Palestinian	42.7	**42.8**	**62.9**	61.6	**0.82**	0.81
Alexandria	Saudi	**43.8**	42.5	**63.5**	61.6	**0.83**	0.82
Alexandria	Sudanese	**42.1**	40.7	**62.9**	60.8	**0.83**	0.81
Alexandria	Syria	**52.9**	49.8	**69.1**	66.8	**0.84**	0.83
Alexandria	Tunisian	**50.2**	48.6	**69.4**	66.5	**0.84**	0.82
Alexandria	Yemeni	**37.0**	36.7	**57.9**	56.6	**0.81**	0.80
APT	Egyptian	2.0	**21.8**	24.7	**48.6**	**0.73**	**0.73**
APT	Levantine	14.8	**15.8**	**44.7**	43.8	**0.72**	0.71
Bible	Moroccan	**28.6**	20.4	**49.7**	42.2	**0.75**	0.72
Bible	MSA	**32.1**	26.4	**51.9**	47.3	**0.77**	0.76
Bible	MSA	**26.2**	22.6	**47.3**	43.7	**0.76**	0.74
Bible	Tunisian	**27.5**	19.7	**49.0**	42.4	**0.74**	0.71
FLORES	Egyptian	**33.9**	31.1	**61.0**	57.5	**0.82**	0.80
FLORES	Iraqi	**36.3**	33.6	**63.6**	59.8	**0.83**	0.81
FLORES	Moroccan	**33.2**	29.9	**59.1**	55.5	**0.80**	0.78
FLORES	MSA	**44.6**	38.3	**68.7**	63.8	**0.85**	0.83
FLORES	N. Levantine	**41.9**	37.8	**66.6**	61.7	**0.83**	0.81
FLORES	Najdi	**42.1**	37.9	**67.2**	63.5	**0.83**	0.82
FLORES	S. Levantine	**45.8**	42.2	**69.2**	64.9	**0.84**	0.82
FLORES	Tunisian	**34.5**	30.6	**61.2**	57.1	**0.81**	0.80
FLORES	Yemeni	**38.8**	36.0	**65.3**	61.7	**0.83**	0.82
MADAR	Egyptian	**47.6**	47.3	**65.9**	65.2	**0.85**	**0.85**
MADAR	Lebanese	**51.1**	47.2	**67.2**	63.8	**0.85**	0.84
MADAR	Moroccan	**46.7**	45.4	**64.3**	62.8	**0.84**	0.83
MADAR	MSA	**54.2**	51.8	**70.5**	68.4	**0.88**	0.86
MADAR	Qatari	**53.0**	47.3	**68.7**	64.2	**0.87**	0.85
MADAR	Tunisian	**44.1**	41.0	**61.5**	58.5	**0.83**	0.81
QAraC	Qatari	**20.2**	19.3	**48.8**	46.1	**0.74**	0.72
Average		**38.4**	35.8	**60.0**	57.4	**0.81**	0.80

Table 17: Gemini and Jais results on the diacritized Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

G.2 Dotless Input

Corpus	Dialect	BLEU				chrF++				COMET			
		Base	Vocab+	Byte	Pixel	Base	Vocab+	Byte	Pixel	Base	Vocab+	Byte	Pixel
Alexandria	Egyptian	4.0	1.4	3.1	**16.2**	19.5	15.9	20.0	**36.7**	0.50	0.44	0.50	**0.65**
Alexandria	Jordanian	5.5	2.5	4.0	**18.8**	20.9	16.3	20.4	**37.8**	0.51	0.45	0.51	**0.66**
Alexandria	Lebanese	5.0	2.0	3.2	**17.8**	21.5	17.1	20.2	**37.4**	0.51	0.45	0.50	**0.64**
Alexandria	Libyan	3.5	1.6	2.8	**13.4**	18.6	14.8	19.3	**33.9**	0.49	0.42	0.50	**0.63**
Alexandria	Mauritanian	3.7	1.8	2.5	**10.0**	20.0	16.2	19.2	**28.8**	0.50	0.44	0.49	**0.58**
Alexandria	Moroccan	3.8	2.0	3.2	**10.8**	18.8	15.9	19.2	**30.0**	0.49	0.43	0.49	**0.59**
Alexandria	Omani	3.9	1.9	3.3	**16.5**	19.5	16.0	19.9	**37.1**	0.50	0.44	0.51	**0.66**
Alexandria	Palestinian	4.9	2.6	3.8	**18.9**	20.1	16.1	20.0	**38.4**	0.50	0.44	0.50	**0.65**
Alexandria	Saudi	4.9	2.2	4.3	**18.2**	20.0	16.1	20.4	**38.3**	0.50	0.43	0.51	**0.66**
Alexandria	Sudanese	4.4	1.7	3.8	**17.4**	20.3	15.5	20.5	**37.5**	0.50	0.43	0.50	**0.64**
Alexandria	Syrian	4.9	2.5	4.4	**23.0**	21.1	17.0	20.9	**42.0**	0.51	0.45	0.51	**0.68**
Alexandria	Tunisian	4.7	2.1	3.3	**13.7**	21.6	17.2	20.4	**34.3**	0.51	0.45	0.50	**0.61**
Alexandria	Yemeni	5.0	2.8	3.9	**15.7**	20.8	17.6	20.7	**35.9**	0.51	0.45	0.51	**0.65**
APT	Egyptian	2.3	1.0	1.8	**9.3**	18.3	14.8	17.7	**30.8**	0.49	0.42	0.48	**0.58**
APT	Levantine	2.0	0.8	1.9	**9.6**	18.7	14.7	18.1	**31.9**	0.49	0.43	0.49	**0.60**
Bible	Moroccan	1.1	0.5	0.8	**3.1**	15.8	14.0	16.4	**22.3**	0.43	0.38	0.43	**0.51**
Bible	MSA	2.1	1.1	1.6	**8.0**	17.4	14.3	17.3	**27.9**	0.45	0.39	0.45	**0.58**
Bible	MSA	1.4	0.9	1.2	**5.1**	15.4	13.9	16.0	**23.4**	0.44	0.38	0.44	**0.54**
Bible	Tunisian	1.4	0.9	1.0	**4.5**	15.5	13.3	16.2	**22.9**	0.43	0.38	0.43	**0.52**
FLORES	Egyptian	1.6	0.7	0.9	**10.7**	17.7	12.8	17.6	**35.1**	0.43	0.36	0.44	**0.63**
FLORES	Iraqi	1.8	0.9	1.5	**11.8**	17.7	13.4	18.0	**36.0**	0.44	0.37	0.44	**0.63**
FLORES	Moroccan	1.3	0.7	0.8	**7.9**	16.6	12.4	17.7	**30.6**	0.42	0.36	0.43	**0.57**
FLORES	MSA	1.8	0.9	1.3	**15.0**	18.1	13.6	18.3	**39.3**	0.44	0.37	0.44	**0.65**
FLORES	N. Levantine	1.9	1.0	1.5	**12.3**	17.3	13.1	18.3	**36.0**	0.43	0.37	0.44	**0.62**
FLORES	Najdi	1.8	0.8	1.4	**14.6**	17.9	13.3	18.3	**39.1**	0.44	0.37	0.44	**0.64**
FLORES	S. Levantine	2.0	1.2	1.7	**14.7**	18.1	13.7	18.9	**38.1**	0.44	0.37	0.45	**0.62**
FLORES	Tunisian	1.6	0.8	1.1	**10.1**	17.3	13.4	17.9	**34.1**	0.43	0.37	0.44	**0.61**
FLORES	Yemeni	1.7	0.8	1.4	**13.0**	17.8	13.2	18.3	**37.7**	0.44	0.37	0.44	**0.64**
MADAR	Egyptian	4.4	2.1	3.3	**15.5**	19.0	15.3	18.1	**34.9**	0.51	0.46	0.50	**0.64**
MADAR	Lebanese	4.6	2.2	2.7	**16.0**	18.7	15.1	17.3	**33.8**	0.52	0.45	0.50	**0.63**
MADAR	Moroccan	3.8	1.4	2.5	**11.4**	18.5	14.8	17.6	**29.6**	0.50	0.44	0.49	**0.60**
MADAR	MSA	5.8	2.8	4.3	**24.7**	22.0	17.8	20.1	**44.0**	0.53	0.47	0.51	**0.69**
MADAR	Qatari	6.6	2.2	4.3	**19.1**	21.0	15.9	19.4	**36.7**	0.54	0.46	0.51	**0.65**
MADAR	Tunisian	3.7	1.1	3.0	**11.5**	17.3	13.9	16.8	**29.1**	0.50	0.44	0.49	**0.60**
QAraC	Qatari	2.3	0.9	2.0	**7.6**	16.8	12.7	16.5	**27.6**	0.49	0.42	0.48	**0.56**
Average		3.3	1.5	2.5	**13.3**	18.7	14.9	18.6	**34.0**	0.48	0.42	0.48	**0.62**

Table 18: The DA+MSA$_{1:1}$ fine-tuned models results on the dotless Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.

		BLEU		chrF++		COMET	
Corpus	**Dialect**	**Gemini**	**Jais**	**Gemini**	**Jais**	**Gemini**	**Jais**
Alexandria	Egyptian	**36.2**	15.0	**58.3**	35.9	**0.81**	0.67
Alexandria	Jordanian	**38.5**	14.1	**58.7**	34.2	**0.81**	0.66
Alexandria	Lebanese	**39.4**	16.8	**59.4**	35.7	**0.81**	0.65
Alexandria	Libyan	**25.2**	7.8	**48.0**	27.2	**0.76**	0.58
Alexandria	Mauritanian	**33.8**	10.7	**54.1**	30.6	**0.78**	0.62
Alexandria	Moroccan	**28.0**	7.0	**48.2**	26.5	**0.74**	0.58
Alexandria	Omani	**34.0**	10.9	**55.9**	32.1	**0.80**	0.64
Alexandria	Palestinian	**39.5**	15.1	**60.2**	34.5	**0.81**	0.65
Alexandria	Saudi	**41.4**	15.8	**61.1**	36.0	**0.82**	0.66
Alexandria	Sudanese	**36.1**	10.9	**57.7**	30.3	**0.79**	0.59
Alexandria	Syrian	**47.5**	17.2	**65.1**	36.9	**0.83**	0.65
Alexandria	Tunisian	**44.9**	16.5	**65.4**	38.0	**0.83**	0.67
Alexandria	Yemeni	**33.8**	13.9	**55.0**	33.4	**0.80**	0.65
APT	Egyptian	1.7	**6.3**	22.7	**25.3**	**0.69**	0.55
APT	Levantine	**12.8**	3.7	**40.8**	23.1	**0.68**	0.54
Bible	Moroccan	**26.1**	8.5	**47.4**	27.4	**0.74**	0.58
Bible	MSA	**31.0**	12.1	**50.7**	31.1	**0.76**	0.61
Bible	MSA	**25.4**	9.6	**46.4**	27.5	**0.74**	0.59
Bible	Tunisian	**26.3**	6.8	**47.1**	24.9	**0.73**	0.56
FLORES	Egyptian	**27.9**	8.4	**55.8**	31.2	**0.79**	0.59
FLORES	Iraqi	**28.1**	6.7	**56.3**	29.5	**0.79**	0.58
FLORES	Moroccan	**25.8**	5.4	**52.2**	27.1	**0.76**	0.55
FLORES	MSA	**38.1**	10.7	**63.9**	33.8	**0.83**	0.61
FLORES	N. Levantine	**34.8**	7.6	**60.5**	29.9	**0.80**	0.58
FLORES	Najdi	**34.6**	8.1	**61.5**	31.6	**0.81**	0.59
FLORES	S. Levantine	**38.0**	9.5	**62.6**	32.1	**0.81**	0.58
FLORES	Tunisian	**27.2**	7.0	**54.5**	28.8	**0.78**	0.57
FLORES	Yemeni	**30.9**	8.6	**58.4**	31.3	**0.80**	0.59
MADAR	Egyptian	**43.2**	14.3	**61.5**	32.3	**0.82**	0.63
MADAR	Lebanese	**44.1**	11.5	**60.0**	28.3	**0.81**	0.60
MADAR	Moroccan	**38.4**	12.2	**56.7**	30.2	**0.80**	0.62
MADAR	MSA	**50.5**	18.9	**67.4**	37.4	**0.86**	0.66
MADAR	Qatari	**43.4**	15.6	**59.5**	32.3	**0.81**	0.62
MADAR	Tunisian	**35.5**	9.2	**52.9**	25.0	**0.78**	0.58
QAraC	Qatari	**15.5**	3.5	**42.0**	21.4	**0.68**	0.54
Average		**33.1**	10.7	**55.1**	30.7	**0.78**	0.61

Table 19: Gemini and Jais results on the dotless Test sets across the dialects for each dataset in terms of BLEU, chrF++, and COMET. Best results are in bold.